

5
H.W: Trace Quick sort as shown in the class for the following array

1) 85 76 99 13 45 103 -5 26 31

2) 350 790 400 200 100 96 900 890 7

sol:- first-call:- $qs(a, 0, 8)$

$n=9$ first=0 last=8 pivot=0

i	j	a[i]	a[j]	a[pivot]
0	8	85	31	85
1	8	76	31	85
2	8	31	99	85
3	8	13	99	85
4	8	45	99	85
5	8	103	99	85
5	7	103	26	85
5	7	26	103	85

85 | 76 | 99 | 13 | 45 | 103 | -5 | 26 | 31

i ↑ j

85 | 76 | 31 | 13 | 45 | 103 | -5 | 26 | 99

i j

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2	8	31	99	85
3	8	13	99	85
4	8	45	99	85
5	8	103	99	85
5	7	103	26	85
5	7	26	103	85

85 | 76 | 99 | 13 | 45 | 103 | -5 | 26 | 31

i ↑ j

85 | 76 | 31 | 13 | 45 | 103 | -5 | 26 | 99

i j

85	76	31	13	45	26	-5	103	99
----	----	----	----	----	----	----	-----	----

$\swarrow$ $\swarrow$ $\swarrow$
 j i $a[j] > pivot$

i	j	a[i]	a[j]	pivot
6	7	-5	103	85
7	7	103	103	85
7	7	103	103	

j and i are crossed then

$$temp = a[pivot] = j$$

$$a[pivot] = a[j]$$

$$a[j] = temp$$

-5	76	31	13	45	26	85	103	99
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-5	76	31	13	45	26
----	----	----	----	----	----

$\swarrow$ $\swarrow$ $\swarrow$ $\swarrow$ $\swarrow$
 j i $a[i] > pivot$

103	99
-----	----

$\swarrow$ $\swarrow$
 j i

$n=6$, first=0, last=5, pivot=0

i	j	a[i]	a[j]	pivot
0	5	-5	26	-5
1	5	76	26	-5
1	5	76	26	-5
1	4	76	45	-5
1	3	76	13	-5
1	2	76	31	-5
1	1	76	76	-5
1	0	76	-5	-5

$76 < -5$ X

$26 > -5$

$45 > -5$

$13 > -5$

$31 > -5$

$76 > -5$

$-5 > -5$ X

$n=2$ first = 7 last = 8 pivot = 7

i	j	a[i]	a[j]	pivot
7	8	103	99	103 $103 < 103$
8	8	99	99	103 $99 > 103$
8	8	99	103	99 $99 < 103$

$\boxed{103} \quad 99$
 $\boxed{99} \quad 103$
 $\boxed{99} \quad \boxed{103}$

swap and

partition $q5(a, 1, 5)$

(-5) (76, 31, 13, 45, 26) first = 1, last = 5

pivot = 76

i	j	a[i]	a[j]	pivot
1	5	76	26	76 $76 > 26$
2	5	31	26	76 $31 < 76$
3	5	13	26	76 $13 < 76$
4	5	45	26	76 $45 < 76$
5	5	26	26	76 $26 > 76$

$\boxed{76} \quad \boxed{31} \quad \boxed{13} \quad \boxed{45} \quad \boxed{26}$
 $\boxed{26} \quad \boxed{31} \quad \boxed{13} \quad \boxed{45} \quad \boxed{76}$

temp = a[5], 76 = 26, 26 = temp.
partition.

$\boxed{26} \quad \boxed{31} \quad \boxed{13} \quad \boxed{45} \quad \boxed{76}$

$\boxed{26} \quad \boxed{31} \quad \boxed{13} \quad \boxed{45} \quad \boxed{76}$

$q5(a, 1, 4)$

first = 1, last = 4, pivot = 26

i	j	a[i]	a[j]	pivot
1	4	26	45	26
2	4	31	45	26 $31 > 26$
3	4	13	45	26 $13 < 26$
4	4	45	31	26 $45 > 26$
4	4	45	31	26 $45 > 26$

$\boxed{26} \quad \boxed{31} \quad \boxed{13} \quad \boxed{45}$
 $\boxed{26} \quad \boxed{31} \quad \boxed{13} \quad \boxed{45}$
 $\boxed{26} \quad \boxed{31} \quad \boxed{13} \quad \boxed{45}$

i and j are crossed

(45, 31, 13, 26)

26

45 | 31 | 13

first = 1, last = 3, pivot = 1

qs(a, 1, 3)

45 | 31 | 13

i j

i	j	a[i]	a[j]	pivot
1	3	45	13	45 45 < 13 ✓
2	3	31	13	45 31 < 45 ✓
3	3	13	13	45 13 > 45 ✗
3	3	45	45	45

temp = a[pivot]

a[pivot] = a[j]

a[j] = a[pivot]

13 | 31 | 45

(2, 3)
(13, 31)

first = 2, last = 3, pivot = 2

13 | 31

i	j	a[i]	a[j]	pivot
2	3	13	31	13 13 < 31 ✓
3	3	31	31	13 31 > 13 ✗

13

31

2)

350	790	400	200	100	96	900	890	760
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$\begin{matrix} 0 & 1 & & & & 5 & 6 & 7 & 8 \\ \swarrow & & & & & \swarrow & \swarrow & \swarrow & \swarrow \end{matrix}$
 first = 0, last = 8, pivot = 350

i	j	a[i]	a[j]	Pivot
0	8	350	760	350
1	8	790	760	350 $790 > 350$
1	8	790	760	350 $790 > 350$
1	7	790	900	350
1	6	790	900	350

swap 790 $\rightarrow$ 96

350	96	400	200	100	790	900	890	760
-----	----	-----	-----	-----	-----	-----	-----	-----

2	6	96	900	350 $96 < 350$
3	5	400	790	350
2	4	400	100	350

swap 400 $\rightarrow$ 100

350	96	100	200	400	790	900	890	760
-----	----	-----	-----	-----	-----	-----	-----	-----

2	4	100	400	350
3	4	200	400	350
4	4	400	400	350 $400 > 350$
4	4	400	400	350 $400 > 350$

$i > j$
 temp = a[j] ; a[pivot] = a[j] ; a[j] = temp

Array be come

(200, 96, 100) 350 (400, 790, 800, 890, 760)

first = 0, last = 2, pivot = 0

i	j	a[i]	a[j]	Pivot
0	2	200	100	200
1	2	96	100	200 $96 < 200$
2	2	100	100	200 $100 > 200$

96 (9, 0, 2)

200	96	100
-----	----	-----

swap (pivot, a[j])

(100, 96) 200

q(0, 9)

100	96
-----	----

i	j	a[i]	a[j]	PIVOT
0	1	100	96	100
1	1	96	96	160

swap (pivot, a[j])

(96, 100)

q(9, 4, 8)

Now take e.

400	790	900	890	760
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first = 4, last = 8, pivot = 4

i	j	a[i]	a[j]	a(pivot)
4	8	400	760	400
5	8	790	760	400
5	7	790	890	400
5	6	890	900	400
5	5	790	790	400

swap (pivot, a[j])

(400, 790)

790	400	900	890	760
-----	-----	-----	-----	-----

790

(900, 890, 760)

900	890	760
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q(9, 6, 8)

i	j	a[i]	a[j]	PIVOT
6	8	900	760	900
7	8	890	760	900
8	8	900	760	900

swap (PIVOT, a[j])

900 760

$(760, 890, 900)$
 $\downarrow$
 $(760, 890)$

$9,5 (a, b, c)$

i	j	a[i]	a[j]	Pivot
6	7	760	890	760
7	7	890	890	760

$\boxed{760 \quad 890}$

$\begin{matrix} \times & j \\ j & i \end{matrix}$